Deep Learning for Medical Image Classification

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1 Introduction and Project Objective

The goal of this project is to design and develop a Deep Learning-based system capable of analyzing chest X-ray images and automatically classifying them into two categories: **NORMAL** and **PNEUMONIA**. Using neural network architectures, the system aims to assist in the detection of pneumonia from radiographic images, providing a modern tool for medical image classification.

Convolutional Neural Networks (CNNs) were quickly identified as the most suitable approach for this task due to their ability to automatically learn relevant features directly from image data, eliminating the need for extensive manual feature extraction. This makes them particularly effective for image classification problems, including medical imaging applications.

In the context of chest X-ray analysis, CNNs can learn to recognize subtle visual patterns such as texture changes, contrast variations, and structural abnormalities associated with pneumonia.

2 Architectural Selection

During the exploratory phase of the project, a comparative evaluation was conducted between the two most prominent architectures for this type of works: **ResNet34** and **DenseNet121**. Both networks were tested using the same transfer learning pipeline to determine which was the best in feature extraction capabilities for chest radiographs.

From the very first experiments, **DenseNet121** consistently achieved better results than the other model, showing higher accuracy and more stable convergence. Also, the architecture was used in the original study that introduced the chest X-ray dataset, making it easier to compare results with existing research. For these reasons, it was chosen as the main architecture, and further efforts were focused on fine-tuning its hyperparameters and training strategy to maximize performance.

3 Database Analysis

A significant part of the project focused on helping the network identify the same visual cues that radiologists typically rely on when examining chest X-rays. The goal was to emphasize the features that distinguish a healthy lung from one affected by pneumonia, making these patterns more apparent and easier for the model to learn during training.

Several "external" elements unrelated to pneumonia were present and could potentially interfere with the model's predictions. In many cases, these consisted of medical devices or metallic implants used to treat other conditions. Such artifacts can introduce unwanted correlations, making it important to ensure that the model focuses on the lung features rather than external medical equipment.
In the example shown (first image below), the thin white line visible in the upper central region is likely part of a respiratory support tube used to assist a critically ill patient. Importantly, these elements cannot simply be removed, since they are part of the clinical reality and often correlate with the patient's condition. In some cases, they may even carry indirect information about disease severity, so they inevitably become part of what the model learns from the data. On the other side the network could also take it as a shortcut and categorize every patient with such tube as ill. This aspect remains a key weak point. Besides potential disturbing factors, The key aspect in recognizing the diseased lungs is *the significant pathological opacification where normal air spaces are replaced by fluid or mucus* resulting in dense white areas or consolidation that obscures the normal anatomy of the lungs.

Ultimately, this work focused predominantly on training and fine-tuning the CNN to accurately map and identify this key feature, leveraging the contrast **between the blurred, opacified areas of disease and the sharp, clear structures of healthy tissue.**

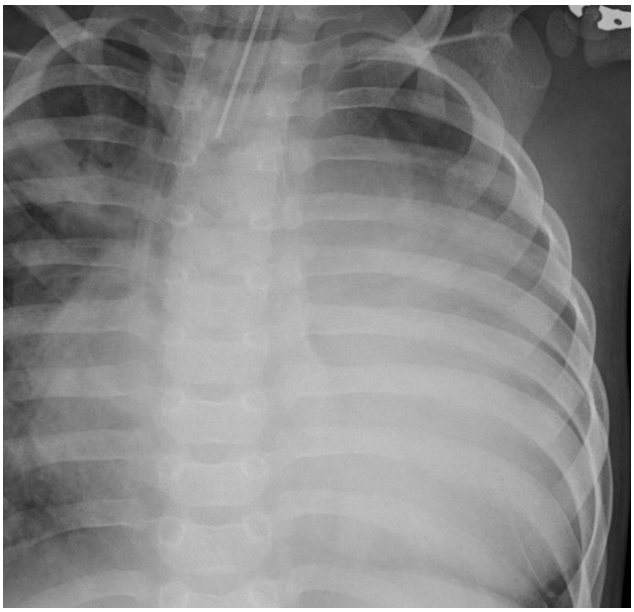


Figure 1: Tube presence in an ill patient's trachea

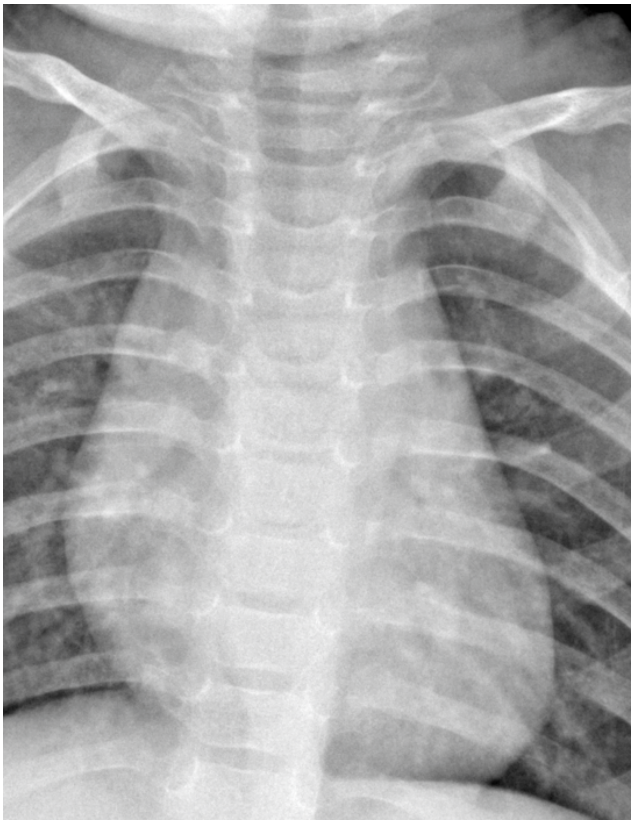


Figure 2: Healthy lungs with defined fields that indicate proper aeration

4 Data Manipulation

Another key aspect is that the dataset exhibits a big class imbalance, where approximately 74.22% of the samples belong to the PNEUMONIA class and 25.78% belong to the NORMAL class.
When a convolutional neural network is trained on such an unbalanced distribution, it tends to become biased toward the majority class. In this case, the model could achieve apparently good accuracy simply by predicting **PNEUMONIA** most of the time, while performing poorly on correctly identifying **NORMAL** cases.

To fix the imbalance the first solution was implementing a `WeightedRandomSampler`: to balance the classes at the batch level by oversampling the minority class. However, this approach alone wasn't enough. Because the pool of **NORMAL** images was significantly smaller, the sampler repeatedly fed the same healthy radiographs to the network within each epoch, so they are not as "helpful" as new images to the network.
Moreover, we can also assume that for the network it is more difficult to grasp what makes a lung "healthy" than to spot clear signs of disease. In this dataset, the **PNEUMONIA** class is quite easy to spot for a model like the used DenseNet, which learns to recognize these obvious signs almost instantly, hitting nearly 99% confidence after just a few runs.
The catch is that under the standard loss function, even when the model is highly confident, those thousands of easy cases still generate a tiny error. When that tiny error gets multiplied by thousands of images, it adds up to a huge number that completely dominates the training process.
This leaves a blind spot for the **NORMAL** class. Healthy X-rays are rarely perfectly clear; they tend to have confusing elements like rib cage shadows, poor lighting or foreign objects as shown previously. Without a way to silence the "noise" from the easy pneumonia cases, the model has no real reason to learn these subtle, borderline differences. Instead, whenever it sees a blurry or ambiguous area, it simply guesses **PNEUMONIA**.

To break this habit, what is called **Focal Loss** was added. By design, whenever the network is certain about a correct guess, the loss for that sample drops to almost zero and is basically ignored. However, if the model is struggling or unsure, the full weight of the mistake hits the network. This forces the network to stop over-analyzing the obvious cases and shifts the majority of its focus to the "gray areas." As a result, the model should be compelled to study the more subtle details of healthy lungs, breaking its bias toward the majority class.

However from a clinical standpoint, it's still better to mistake a healthy person for a sick one than the other way around. The model should always prioritize **Sensitivity** (Recall), because missing a positive case (a False Negative) can have serious consequences. This point was taken into account during the fine-tuning phase.

To evaluate the neural network fairly, a validation set was added by subdividing the training set into two separate subsets using an 80/20 ratio. This resulted in 1,047 validation images that were kept completely separate from the 624 test images.
Maintaining a strict separation between the training, validation, and test sets is essential

because it prevents information leakage. As a result, the final performance metrics, such as Accuracy and Recall, provide a reliable estimate of how well the model is expected to perform on previously unseen patient X-ray images in real-world clinical settings.

5 Transfer Learning

Training a deep convolutional neural network from scratch typically requires a very large collection of labeled medical images to achieve good performance. Since obtaining such datasets can be challenging, this project leverages transfer learning by using the pre-trained DenseNet model as the backbone, as explained at the beginning. To adapt the network for the binary classification task of distinguishing between Normal and Pneumonia chest X-ray images, the original classification layer trained on the ImageNet dataset was removed and replaced with a custom classification head. The original ImageNet classifier was designed to recognize over 1,000 different object categories, making it unsuitable for this much simpler two-class problem.

A very common approach with this type of datasets is keeping initially the backbone frozen so that only the newly added classification head is trained. This means the model initially learns just how to interpret the high-level features, rather than modifying the full set of weights.
This approach works because the early and intermediate layers of convolutional networks tend to learn very general visual patterns, such as edges, textures, shapes, and simple structures. These kinds of features are not specific to one dataset and can be effectively reused in very different tasks, including medical image analysis.
Starting from the third epoch, the backbone is unfrozen to further adapt the model to the complexity of chest X-ray data. This controlled process avoids destabilizing the training, which can happen if all layers start updating simultaneously.

Furthermore, once the backbone is unfrozen, its layers are not updated at the same rate. To account for this, the model parameters are divided into three groups:

- early layers,
- late layers,
- classification head.

The optimizer is then configured with a different learning rate for each group.
This matters because early layers already know how to spot basic things like edges and textures, which are the same whether you're looking at a cat or a lung. They just need a tiny fine-tuning with a very low learning rate. On the other hand, deep layers were originally trained to recognize complex everyday objects, so they need to adapt much faster using a higher learning rate to learn completely new medical patterns, like lung anatomy and pneumonia lesions

6 Data Augmentation

Models like DenseNet were originally trained on ImageNet, where images are standardized to 224 × 224 pixels. This creates a mismatch in medical imaging, since chest X-rays and similar scans are typically captured at much higher resolutions. If these images are directly fed into the network without a proper preprocessing strategy, important fine details—such as small nodules or subtle opacities—could be lost or distorted during resizing.

To address this, the images are processed in a way that preserves as much information as possible before they are passed to the model.

- Each image is resized while maintaining its original aspect ratio, ensuring that anatomical structures are not stretched or compressed.
- The image is then padded to form a square input while preserving its original aspect ratio. Instead of adding black bands around the image, which would create unnatural contrasts at the borders, the padding is generated by reflecting the neighboring X-ray pixels.
- When the image must be reduced in size, interpolation is used to minimize artifacts and retain fine details, as it doesn't just remove the additional underneath pixels but returns an average.

This approach helps preserve the visual characteristics that are important for accurate medical image analysis.

In order to try revealing hidden details in medical imaging, a **contrast enhancer** was also applied. In chest X-rays, critical diagnostic indicators like pneumonia often present as faint, low-contrast opacities that can easily blend into surrounding structures.

For this purpose **CLAHE** (Contrast Limited Adaptive Histogram Equalization) was chosen. Instead of changing the contrast of the entire image, CLAHE splits the image into small regions and improves the contrast in each region independently. This makes subtle tissue abnormalities and lesions easier to see while avoiding overexposure of dense structures like the ribs and spine.

In the provided architecture, It's configured with a probability of **p=0.5** exclusively within the training dataset. Applying this enhancement to only 50% of the training images instead of on 100% of the whole batch, forces the neural network to learn how to identify pathological features under two completely different visual conditions: once when the boundaries are artificially sharpened and enhanced, and once when the image remains raw. This variance prevents the model from developing a strict dependency on high-contrast visuals, effectively reducing overfitting and increasing its capacity to generalize across unpredictable, real-world clinical environments.

CLAHE is omitted from the validation and test sets ('p=0') to ensure a deterministic eval-

uation. This setup guarantees that the model is tested against standard, unaltered medical scans exactly as they are captured by radiology departments.

7 Results

The trained network showed a good overall performance when evaluated on the independent test set. Its *ROC-AUC* score reached 0.9607, indicating an excellent ability to distinguish between healthy lungs and lungs affected by pneumonia across a wide range of decision thresholds. This result suggests that the model is capable of reliably separating the two classes and generalizing well to previously unseen data.
An analysis of the training process revealed that the model converged quickly. By epoch 7-8, both the *validation loss*, the *training loss* and the main diagnostic metrics had optimized to competitive levels with no meaningful improvements observed during subsequent training.
The slight physiological performance drop between the validation set and the independent test set highlights the expected real-world variance, yet confirms the network successfully avoided severe overfitting.

As anticipated, the strong class imbalance in the dataset made it difficult to achieve high *specificity*.
When the decision threshold is increased to 0.7, the model reaches its best overall *accuracy*, providing the most effective balance between correctly identifying pneumonia cases and reducing misclassifications. In a clinical setting, however, not all errors carry the same consequences. Missing a patient with pneumonia (a false negative) can delay treatment and potentially lead to serious health complications, whereas incorrectly classifying a healthy patient as ill (a false positive) generally results in additional examinations or monitoring. For this reason, it is preferable for the model to err on the side of caution and generate more false positives rather than risk overlooking true pneumonia cases. Consequently, a threshold of 0.7 represents not only the optimal trade-off between sensitivity, specificity, and overall accuracy, but also a practical upper limit beyond which the increase in missed positive cases would outweigh any gains in specificity.

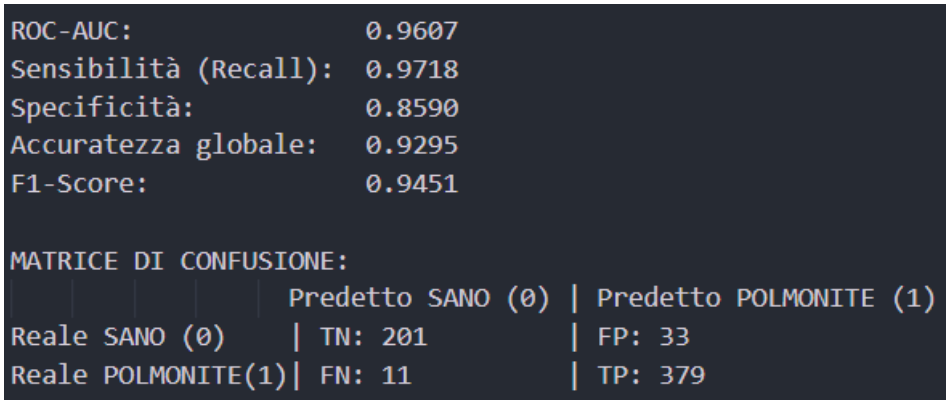


Figure 3: Final Results with a 0.7 threshold

8 Conclusion

In conclusion, this project highlights how neural networks can be thoughtfully tailored to the realities of clinical diagnostics. Rather than just letting a network run wild on raw data, the implementation of targeted data manipulation and a phased, fine-tuned training architecture successfully pushed the model to focus on the actual, subtle signs of disease within the lungs.

From a practical standpoint, the true value of this system lies in its adaptability. By moving away from rigid, standard classification limits and evaluating a spectrum of decision thresholds, the model can be dynamically tuned based on the immediate clinical context, whether the goal is a conservative screening baseline that prioritizes catching every potential illness, or a balanced triage tool that optimizes overall accuracy.